Employee Turnover Analytics

Project Statement:

Portobello Tech is an app innovator that has devised an intelligent way of predicting employee turnover within the company. It periodically evaluates employees' work details including the number of projects they worked upon, average monthly working hours, time spent in the company, promotions in the last 5 years, and salary level.

Data from prior evaluations show the employee's satisfaction at the workplace. The data could be used to identify patterns in work style and their interest to continue to work in the company.

The HR Department owns the data and uses it to predict employee turnover. Employee turnover refers to the total number of workers who leave a company over a certain time period.

As the ML Developer assigned to the HR Department, you have been asked to create ML Programs to

1. Perform data quality check by checking for missing values if any.

```
[356]: df_emp.dtypes.value_counts()
```

```
[356]: object     13
        int64       7
        float64     2
        Name: count, dtype: int64
```

Sol 1. Checking the data quality

```
[358]: df_emp.isna().sum(axis = 0)
```

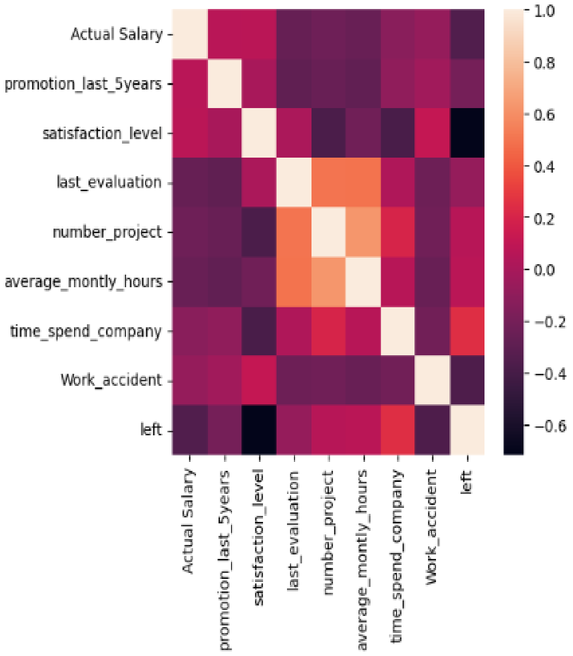
```
[358]: Employee ID                0
        First Name                 0
        Last Name                  0
        Gender                     0
        State                      0
                                 ...
        average_montly_hours       0
        time_spend_company         0
        Work_accident              0
        left                       0
        Termination Date       13315
        Length: 22, dtype: int64
```

2. Understand what factors contributed most to employee turnover by EDA.

```
plt.figure(figsize=(5,5))
sns.heatmap(df_emp_corr.corr(),annot=False)
plt.show()
```

[367]: <Figure size 500x500 with 0 Axes>

[367]: <Axes: >

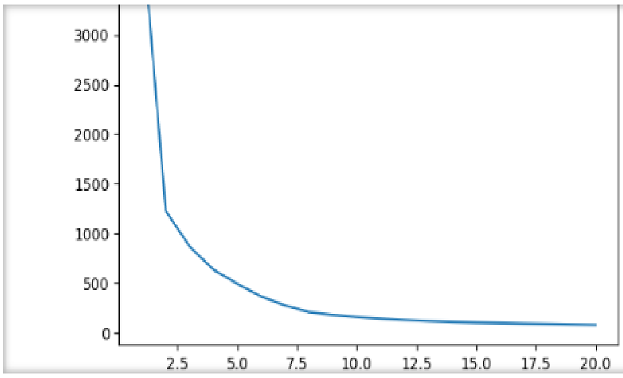


There are many other insights we observed which are provided in Presentation slides to discuss with the Business.

3. Perform clustering of Employees who left based on their satisfaction and evaluation.

```
78.6679088487748]
```

```
[398]: # employing the elbow method to find correct number of clusters by plotting wcss on y and nu. of clusters on x.
plt.plot(range(1,21),wcss)
```



So here we find that number of clusters should be 7 or 8 since at these the rate of change in slope decreases. But lets try with 6 clusters.

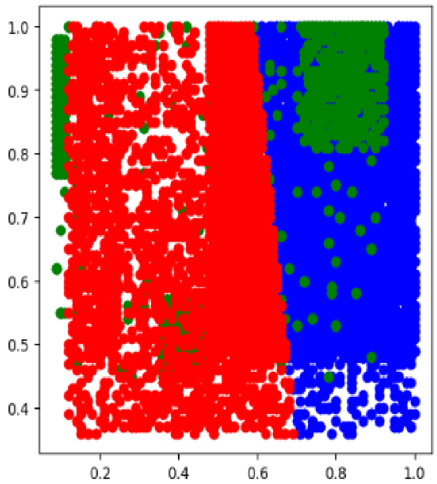
```
[400]: km = df_clus.iloc[:,:].values
km
```

```
[400]: array([[0.38, 0.53, '1'],
       [0.8, 0.86, '1'],
       [0.11, 0.88, '1'],
       ...,
       [0.37, 0.53, '1'],
       [0.11, 0.96, '1'],
       [0.37, 0.52, '1']], dtype=object)
```

```
[401]: from sklearn.cluster import KMeans
```

```
plt.scatter(km[y_means==2,0],km[y_means==2,1],color = 'red')
# plt.scatter(km[y_means==3,0],km[y_means==3,1],color = 'black')
plt.show()
```

```
[405]: <Figure size 500x500 with 0 Axes>
[405]: <matplotlib.collections.PathCollection at 0x2528d9538d0>
[405]: <matplotlib.collections.PathCollection at 0x2528a1568d0>
[405]: <matplotlib.collections.PathCollection at 0x2528d96bed0>
```




```
[406]: plt.scatter(kmeans.cluster_centers_[:, 0], kmeans.cluster_centers_[:, 1],
```

4. Handle the left Class Imbalance using SMOTE technique.

By looking at the value counts it is seen that the dataset is highly imbalanced . So we need to balance it to get unbiased results. Hence we are using SMOTE Technique to balanced the dataset.

Sol 4. Handle the left Class Imbalance using SMOTE technique

```
[421]: import imblearn
       from imblearn.over_sampling import SMOTE
       sm = SMOTE(random_state = 2)
```

```
[422]: x_trains,y_trains = sm.fit_resample(x_train,y_train)
       x_tests,y_tests = sm.fit_resample(x_test,y_test)
```

```
[423]: y_trains.value_counts()
```

```
[423]: left
       0    9134
       1    9134
       Name: count, dtype: int64
```

5. Perform k-fold cross-validation model training and evaluate performance.

```
1    9134
Name: count, dtype: int64
```

Sol 5.Perform k-fold cross-validation model training and evaluate performance

```
[425]: from sklearn.model_selection import cross_val_score
       from sklearn.linear_model import LogisticRegression
       from sklearn.ensemble import RandomForestClassifier
       from sklearn.metrics import roc_auc_score
       import sklearn.metrics as metrics
```

```
[426]: # training the model logistic regression
       logreg = LogisticRegression(solver='lbfgs',max_iter=1000)
       print(cross_val_score(logreg,x_trains,y_trains,cv = 5 , error_score = "raise"))
       print(cross_val_score(logreg,x_trains,y_trains,cv = 5 , error_score = "raise").mean())

       [0.55528188 0.55582923 0.54734537 0.56720504 0.80755543]
       0.6066433913777146
```

```
[427]: # testing
       logreg.fit(x_trains,y_trains)
       y_pred = logreg.predict(x_tests)
       y_pred
```

```
[427]: LogisticRegression
       LogisticRegression(max_iter=1000)
```

```
[427]: array([0, 1, 1, ..., 1, 0, 1], dtype=int64)
```

```
[428]: from sklearn.metrics import classification_report
       metrics.confusion_matrix(y_tests,y_pred)
```

```
[428]: array([[1267, 1027],
```

6. Identify the best model and justify the evaluation metrics used.

0.9880054873383281

6.Identify the best model and justify the evaluation metrics used.

[433]:
```
# RandomForest
rdm = RandomForestClassifier(max_depth = 5)
print(cross_val_score(rdm,x_trains,y_trains,cv = 5).mean())
```

```
0.9180528229491293
```

[434]:
```
rdm.fit(x_trains,y_trains)
y_pred1 = rdm.predict(x_tests)
y_pred1
```

[434]:
```
RandomForestClassifier
RandomForestClassifier(max_depth=5)
```

[434]:
```
array([0, 0, 0, ..., 0, 0, 1], dtype=int64)
```

[435]:
```
fpr,tpr,threshold = metrics.roc_curve(y_tests,y_pred1)
print(fpr)
print(tpr)
print(threshold)
roc_auc = metrics.auc(fpr,tpr)
print(roc_auc)
```

```
[0.         0.05318221 1.        ]
[0.         0.84437663 1.        ]
[2 1 0]
0.8955972101133393
```

7. Suggest various retention strategies for targeted employees.

[438]:
```
array(['Actual Salary', 'promotion_last_5years', 'satisfaction_level',
       'last_evaluation', 'number_project', 'average_montly_hours',
       'time_spend_company', 'Work_accident', 'Gender_Female',
       'Gender_Male', 'salary_high', 'salary_low', 'salary_medium',
       'Department_IT', 'Department_RandD', 'Department_accounting',
       'Department_hr', 'Department_management', 'Department_marketing',
       'Department_product_mng', 'Department_sales', 'Department_support',
       'Department_technical'], dtype=object)
```

[439]:
```
# importanct of features
importance = rdm.feature_importances_
feature_index_by_dec_imp = importance.argsort()[::-1]
for feature in feature_index_by_dec_imp :
    print('{}:{:.2f}%'.format(feature_labels[feature],importance[feature]*100))
```

```
satisfaction_level:25.82%
time_spend_company:20.24%
number_project:17.28%
average_montly_hours:11.44%
last_evaluation:8.58%
Work_accident:4.70%
Gender_Male:3.76%
Gender_Female:3.70%
Actual Salary:1.36%
Department_technical:0.75%
salary_high:0.66%
Department_sales:0.58%
salary_low:0.40%
Department_support:0.23%
Department_hr:0.13%
salary_medium:0.11%
promotion_last_5years:0.08%
Department_accounting:0.06%
```

So we find that satisfaction_level:26.57% , time_spend_company:19.06 , number_project:16.9 , % average_montly_hours:12. , 7% last_evaluation: are top 5 features which are important to retain the employee . By using this insights we can make retension strategy for the employees.9.70